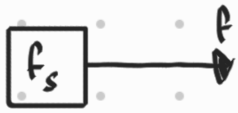
S_e

$e = e_s$

S_f

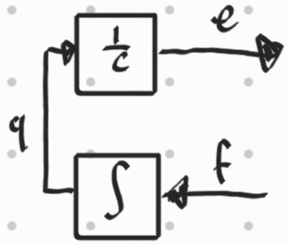


$f = f_s$

C

$e = \frac{q}{C}$

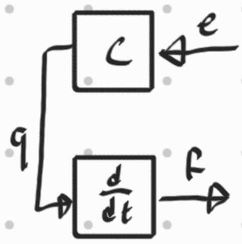
$q = \int f\, dt + p(0)$



C

$q = Ce$

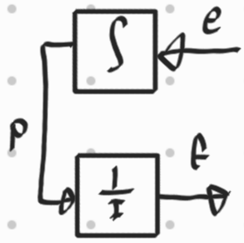
$f = \frac{dq}{dt}$



I

$f = \frac{p}{I}$

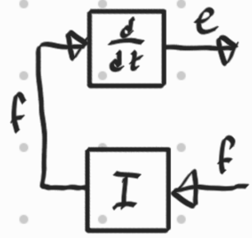
$p = \int e\, dt + p(0)$



I

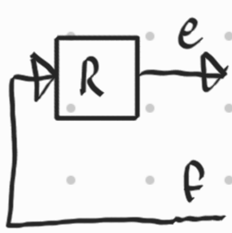
$p = If$

$e = \frac{dp}{dt}$



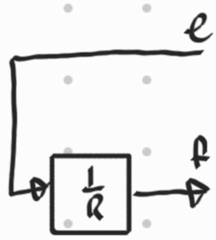
R

$e = Rf$



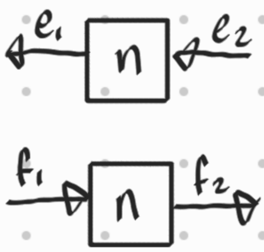
R

$e = \frac{f}{R}$



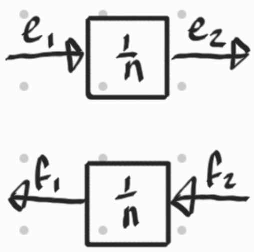
$\overset{e_1}{\underset{f_1}{\rightharpoonup}}\ \underset{n}{TF}\ \overset{e_2}{\underset{f_2}{\rightharpoonup}}$

$f_2 = n f_1$



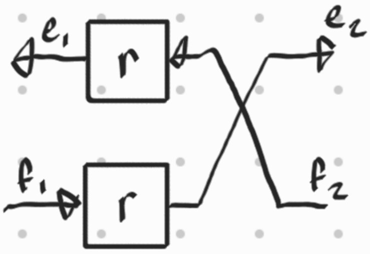
$\underset{n}{TF}$

$f_1 = \frac{f_2}{n}$

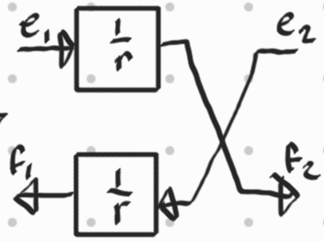


$e_1 = ne_2$ $\qquad$ $e_2 = \frac{\ }{n}$

$$\overset{e_1}{\underset{f_1}{\vdash\!\!\rightharpoonup}}\ \underset{\ddot{r}}{GY}\ \overset{e_2}{\underset{f_2}{\rightharpoonup\!\!\dashv}}$$



$$\rightharpoonup\!\!|\ \underset{\ddot{r}}{GY}\ |\!\!\rightharpoonup$$



$$e_2 = r f_1$$
$$e_1 = r f_2$$

$$f_2 = \frac{e_1}{r}$$
$$f_1 = \frac{e_2}{r}$$

$$\overset{e_R}{\underset{f_R}{\rightharpoonup}}\ RS\ \overset{T}{\underset{f_{S,irr}}{\rightharpoonup}}$$

entropy

$$e_R = R \cdot f_R$$

$$f_{S,irr} = \frac{e_R f_R}{T} = \frac{R \cdot f_R^2}{T} \geqslant 0$$

$$T = \frac{e_R f_R}{f_{S,irr}}$$

$$\overset{T_{in}}{\underset{f_{S_{in}}}{\rightharpoonup}}\ RS\ \overset{T_{out}}{\underset{f_{S_{out}}}{\rightharpoonup}}$$

$$\left.\begin{aligned} f_{S_{in}} &= \frac{dS_{in}}{dt} = \frac{1}{R}\,\frac{(T_{in} - T_{out})}{T_{in}} \\ f_{S_{out}} &= \frac{dS_{out}}{dt} = \frac{1}{R}\,\frac{(T_{in} - T_{out})}{T_{out}} \end{aligned}\right\} T_{S_0} = \frac{1}{R} \cdot \frac{\Delta T}{T_0}$$

$$\begin{pmatrix} e_1 \\ f_1 \end{pmatrix} = \begin{bmatrix} 0 & r \\ 1/r & 0 \end{bmatrix} \begin{pmatrix} e_2 \\ f_2 \end{pmatrix}$$